17 (a) Factorise $3(x - y)^2 - 2(x - y)$

Let $u = x - y$. Then $3u^2 - 2u = u(3u - 2)$.

Common factor $(x - y)$:

1 mark
Factor out(x−y)

$(x - y)[3(x - y) - 2]$

$= (x - y)(3x - 3y - 2)$.

1 mark
(x−y)(3x−3y−2)

(2)

(b) Show that $\dfrac{1}{2x^2 + x - 15} \div \dfrac{1}{3x^2 + 9x}$ simplifies to $\dfrac{ax}{bx + c}$ where a, b and c are integers.

×	$2x$	-5
x	$2x^2$	$-5x$
3	$6x$	-15

$2x^2 + x - 15 = (2x - 5)(x + 3)$

$3x^2 + 9x = 3x(x + 3)$

1 mark
Factor both denominators

$\dfrac{1}{(2x - 5)(x + 3)} \times \dfrac{3x(x + 3)}{1}$

1 mark
Multiply by reciprocal

Cancel the common factor $(x + 3)$:

$\dfrac{3x}{2x - 5}$ $\quad a=3,\ b=2,\ c=-5.$

1 mark
3x/(2x−5)

(3)

(Total for Question 17 is 5 marks)